

izi — XAU Assistant

SECTION 01 // ARCHITECTURE

Autonomous MT5 Execution Engine + AI Forecaster + Telegram Operations

TELEGRAM COMPANION: /manual

izi is a high-performance, institutional-grade XAUUSD (Gold) trading assistant engineered in two resilient halves. The **MQL5 Expert Advisor** (running inside MetaTrader 5 on Windows) executes real-time strategy triggers and manages live baskets. The **Python Engine** (FastAPI in WSL2) grades signals via deep-learning forecasting, streams telemetry to Telegram, and coordinates remote commands.

LAW 1: STRATEGY DECIDES, AI GRADES

The MQL5 EA is the **sole decision maker**. On closed bar, the EA evaluates entry criteria and executes *first* in AUTO mode, then queries AI forecasting. Trading decisions never wait on inference latency.

LAW 2: FAIL-OPEN RESILIENCE

Strict fail-open design across all layers: if the AI service or network is offline, strategy signals continue to alert and execute automatically, flagged '*AI unavailable*'. Service downtime never blocks a trade.

SYSTEM TOPOLOGY & INTEGRATION MATRIX

Component	Environment / Host	Core Responsibility	Port / Location
MQL5 EA	Windows (MT5 Terminal)	Closed-bar signal evaluation, AUTO order execution, basket trail & ratchet management	XauAssistant.mq5
FastAPI Engine	WSL2 (Python 3.13 venv)	Chronos AI signal grading, news event blackout windows, web mini-app, remote router	http://127.0.0.1:9000
SQLite DB	WSL2 Filesystem	Zero-loss audit store: ticks, candles, strategy proposals, AI verdicts & trade logs	xau_assistant.db
Telegram Bot	Cloud API / Webhooks	Instant push alerts, interactive action buttons ([BUY], [SELL], [EXIT]), mini-app charts	@izi_assistant_bot

Bi-directional Heartbeat Pipeline

The MQL5 EA heartbeats the FastAPI service every **5 seconds**. Remote operator actions (manual trades, basket exits, stop-loss ratchets) queue in memory and ride back on the immediate next heartbeat payload. Tap latency to MT5 terminal is ≤ 5 seconds.

Every-Bar Lazy Resolution

The EA sends candle data to /analyze on every closed bar (including **NONE** signals). This allows the service to lazily evaluate outcomes of past signals without maintaining a separate data feed, logging full statistical telemetry into SQLite.

Core Architectural Rules: (1) No martingale: position size never increases after a loss; pyramiding adds strictly into winning legs. (2) Risk state survives crashes via MT5 Global Variables. (3) AI confidence grading is logged from tick zero to ensure empirical evaluation.

Client Setup & Configuration

SECTION 02 // SETUP & PARAMS

Idempotent Single-Script Deployment & Terminal Integration

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PHASE 1: AUTOMATED ZERO-TOUCH SETUP

A single unified script configures the entire stack. Safe to re-run at any time (idempotent; skips completed phases):

```
cd && scripts/setup.sh
```

• **Automated Actions:** Provisions Python venv, installs core dependencies, validates 600+ test suites, starts background service, pairs Telegram bot (message your bot once when prompted), copies MQL5 EA to MT5 data path and compiles it.

PHASE 2: SERVICE CONFIGURATION (.env)

```
cd service && cp .env.example .env
```

Variable	Default / Options	Description & Operational Impact
FORECASTER	chronos fake	chronos loads real neural weights. fake uses fast linear extrapolation for headless testing.
PORT / HOST	9000 / 127.0.0.1	Service port and localhost binding. Keep 127.0.0.1 to avoid exposing credentials to local LAN.
TELEGRAM_BOT_TOKEN	token string	API token obtained from @BotFather for alerts, commands, and web mini-app charts.

PHASE 3: METATRADER 5 TERMINAL INTEGRATION (MANDATORY)

Step 1: Whitelist WebRequest URL in MT5

Navigate to: **Tools > Options > Expert Advisors**

1. Check the box: 'Allow WebRequest for listed URL'

2. Click green + and add exactly: http://127.0.0.1:9000

(MT5 stores this encrypted; requires manual entry)

Step 2: Attach EA & Enable Algo Trading

1. In **Navigator > Expert Advisors**, drag XauAssistant onto a **XAUUSD M5** chart.

2. In the dialog, check 'Allow Algo Trading'.

3. Click **Algo Trading** button in MT5 top toolbar (ensure icon turns green).

(If chart was already open, remove and re-attach the EA)

PHASE 4: KEY EA INPUT PARAMETERS REFERENCE

Parameter	Default	Function & Optimization Context
ExecutionMode	AUTO	AUTO (executes on closed bar) vs MANUAL (proposes with Telegram buttons).
EntryMode	ADR	ADR (1/3 ADR target + partial trailing) vs FIXED (3R runner + SL ratchet).
RiskPercent	1.0%	Risk per trade basket as % of account balance. Auto-sizes lots via ATR stop distance.
MaxSpreadPoints	40	Spread filter threshold in points. Rejects trade entries during volatility widenings.

CRITICAL SAFETY LOCK: REAL-MONEY LIVE ACCOUNTS

On live accounts (real currency), AUTO mode is strictly locked out by default and will refuse trades until the EA input parameter **AllowLiveTrading = true** is explicitly enabled. This intentional safeguard protects capital until paper trading and AI accuracy telemetry earn live execution.

Daily Operations & Lifecycle

SECTION 03 // OPERATIONS & TELEMETRY

Launch Procedures, Telemetry Checks & Safe Maintenance

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1. SYSTEM STARTUP & ORCHESTRATION

Standard 1-Click Launch: Double-click xau-launch.bat on the Desktop — it automatically initialises WSL2, starts the FastAPI service, and launches MetaTrader 5.

Headless / Manual Startup (WSL2):

```
cd /service
nohup .venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 9000 >> service.log 2>&1 &
```

Note: Initial neural AI forecast on startup may take ~1-2 min for weight loading. Trading is never blocked (fail-open).

2. HEALTH MONITORING & TELEMETRY CHECKS

REST API Endpoint

curl http://127.0.0.1:9000/health

Returns JSON payload confirming database read/write, forecaster readiness, and memory utilization.

Telegram Live Snapshot

Run /status in Telegram.

Line 1 displays EA link health (e.g. EA: connected (2s ago)), active mode, and shield statuses.

3. CONTROLLED SHUTDOWN & STATE SURVIVAL

```
kill -f "uvicorn app.main" # Exit code 144 is standard and clean
```

Persistent Risk State: Critical risk parameters (daily loss brake, drawdown high-water mark, kill switch) are stored in **MT5 Global Variables**, not volatile RAM. They survive service crashes, terminal restarts, and PC reboots seamlessly.

4. MULTI-TIER RISK DEFENSE ARCHITECTURE

Tier	Mechanism & Scope	Trigger & Action
Tier 1: Per-Trade	MoneyWatch dynamic lot sizing + ATR stops	Sizes each order to exact risk %; enforces hard stop-loss on every leg.
Tier 2: Daily Brake	Cumulative daily realized loss brake	Alerts at ≥70% spent; halts all new entries at 100% until 00:00 server time.
Tier 3: Kill Switch	Equity drawdown high-water mark protection	Terminal locks if total equity drops below absolute max drawdown watermark.

5. HOT UPDATES & RECOMPILATION PROTOCOL

- Python Service Updates:** Kill first (foreground pkill), start second (background nohup). Never combine in one command.
- MQL5 EA Updates:** Run scripts/setup.sh — it copies files and triggers MetaEditor compile. MT5 hot-reloads the newly compiled EA automatically upon completion with 0 errors.
- Strategy Changes:** New strategies slot cleanly into mt5/Include/XauAssistant/Strategies/ behind CStrategy without touching execution logic.

Telegram Command Center

SECTION 04 // COMMANDS & MINI-APP

Comprehensive Operator Command Directory & Manual Trading Pipeline

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OPERATOR COMMAND DIRECTORY

Commands respond exclusively to the authorized owner chat. Broadcast channels receive sanitized alerts without financial balances or interactive buttons.

Command	Category	Description & Functionality	Output / Behavior
/status	Telemetry	Complete health snapshot: EA heartbeat, mini-app, news blackout & risk shields.	Status report card
/bal	Telemetry	Account balance, live equity, floating P/L, daily & weekly realized performance.	Financial metrics
/mode	Config	Toggle execution mode (AUTO / MANUAL), target mode (ADR / FIXED), and lane.	Active configuration
/agree	Config	Toggle confirmation filters: Higher-Timeframe bias & EMA-200 trend agreement.	Filter status
/trade	Execution	Request manual order entry menu with live gold quote and [BUY] / [SELL] buttons.	Interactive buttons
/news	Risk	Upcoming high-impact USD economic events and active entry blackout windows.	Economic calendar
/chart	Visuals	Launches Telegram Web Mini-App live chart or renders a static candle snapshot.	Interactive chart
/stats	Analytics	Detailed hit rates, expectancy, and win/loss performance per strategy module.	Performance breakdown
/history	Audit	Audit trail of the last 10 execution events, entries, exits, and SL ratchets.	Recent trade logs
/channel	Config	Link or unlink an external Telegram broadcast channel for public trade signals.	Channel binding
/manual	Docs	Instantly compiles and delivers this operator quick-reference PDF to chat.	PDF document

Manual Trade Pipeline (/trade)

1. Operator Tap:

/trade displays live bid/ask with [BUY] / [SELL] buttons.

2. Multi-Gate Check:

Verifies no active basket, no queued entry, and EA connected.

3. Heartbeat Dispatch:

Next 5s heartbeat routes order to MT5.

4. Basket Trail:

Position managed automatically with target exits.

Telegram Web Mini-App (/chart)

• Real-Time Charts:

Full TradingView lightweight candle charts with HalfTrend + EMA overlays.

• AI Verdict Flags:

Real-time visual pins showing neural forecaster confidence & direction.

• Emergency Exit Button:

1-tap full basket close directly from the Web App interface.

Owner Security & Channel Isolation:

Linked broadcast channels (/channel) receive sanitized trade execution announcements for subscribers. Account balances, equity figures, risk metrics, and interactive buttons are strictly withheld. Only the authenticated owner chat retains remote execution authorization.

Controls, Risk & Troubleshooting

SECTION 05 // BUTTONS & DIAGNOSTICS

Button Behaviors, Protection Shields & Quick Diagnostics Matrix

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INTERACTIVE BUTTON DIRECTORY

Button Label	Context / Trigger	System Action & Safety Rule
[Take trade] / [Skip]	MANUAL mode proposals	Approves or rejects pending strategy setup. Auto-expires safely after 15 minutes.
[EXIT - close trade]	Trade entry notification	Immediately closes all open basket positions at market on next heartbeat.
[Move SL to here]	FIXED-ride target alert	One-way ratchet: tightens stops of all legs to current price. Never loosens.
[Reset brake]	Daily loss brake notice	Overrides daily loss halt after manual operator risk assessment (≥70% spent).

RISK SHIELDS & DIAGNOSTIC MATRIX

Automated Risk Shields • USD News Blackout: Freezes new entries 30 min before and after high-impact releases. Alerts sent 35m ahead. Exits never blocked. • Daily Loss Brake: Halts new entries if daily loss reaches threshold. Prevents tilt. • Spread Gate: Blocks entries during spread spikes (>40 pts).	Quick Diagnostics • EA Disconnected: Check MT5 is open, EA on chart (smiley), Algo Trading enabled, WebRequest URL set. • Indicators Missing: MT5 indicators redraw on next live tick; normal during market closure. • Trade Didn't Fire: Check /status for news blackout, spread gate, or daily loss brake.
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COMMON OPERATOR INCIDENTS & REMEDIATION

Observed Incident	Probable Root Cause	Immediate Remediation Action
EA status disconnected	MT5 terminal restarted or EA detached	Re-attach XauAssistant to XAUUSD chart and confirm smiley icon.
Trade rejected: Live Lock	AllowLiveTrading set to false	In EA Inputs, set AllowLiveTrading = true for real accounts.
Trade rejected: Spread Spike	Market spread exceeds max threshold	Wait for volatility/spread normalization during high-volume sessions.

CRITICAL LOG & STORAGE PATHS

service/service.log	FastAPI service runtime log (auto-rotated at 20 MB with rollover backups).
service/xau_assistant.db	SQLite database containing signals, trade fills, heartbeats and forecaster grades.
MT5 'Experts' Tab	MetaTrader 5 terminal log showing raw tick processing, order tickets, and MQL5 alerts.